Name: Priyanshu Digari
Roll no. : 89

Assignment -06
Image Processing and Pattern Recognition
TCS-361

Q1.) Explain different types of image features: color, texture, shape, and local features with suitable examples.

Ans.) 1. Color Features:
Represent distribution of colors in an image.
Example: Color histogram, RGB/HSV values.
Used in image retrieval (e.g., finding red objects).

2. Texture Features:
Describe surface patterns and spatial

variation.
Example: Smooth, rough, striped textures using GLCM, entropy.
Used in medical imaging (tumor detection).

3. Shape Features:
Describe geometric properties of objects.
Example: Area, perimeter, boundary, circularity.
Used in object recognition (detecting coins, faces).

4. Local Features:
Extract key points from small regions.
Example: SIFT, SURF, ORB features.
Used in image matching, stitching, object detection.

Q2.) Explain Principal Component Analysis (PCA) for dimensionality reduction. Discuss its advantages and

limitations.
Ans.) PCA is a statistical method used to reduce dimensionality by transforming data into a new set of orthogonal variables called principal components.

Steps:
1. Normalize data
2. Compute covariance matrix
3. Find eigenvalues and eigenvectors
4. Sort eigenvectors by decreasing eigenvalues
5. Project data onto top k eigenvectors

Advantages:
- Reduces dimensionality
- Removes redundancy
- Improves computation efficiency
- Helps visualization

Limitations:
- Loss of information
- Assumes linear relationships

- Sensitive to scaling
- Hard to interpret components

Q3.) Elaborate the purpose of morphological processing in images.

Ans.) Morphological processing is used to extract and process shapes in images using structuring elements.

Purpose:
- Remove noise
- Enhance structures
- Extract boundaries
- Fill gaps

Basic operations:
1. Erosion: Shrinks objects
2. Dilation: Expands objects
3. Opening: Removes small objects (erosion + dilation)
4. Closing: Fills holes (dilation + erosion)